Research Statement

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I am an applied environmental economist. I am interested in understanding the propagation of energy and climate shocks through public finances, financial markets and supply chains. Much of my work focuses on how fossil fuel revenues affect the economies of producer countries and, in particular, on how the spending of this revenue affects carbon emissions. Other work has explored the causes and incidence of carbon underpricing, the pricing and insurance of climate risk in financial markets, and the impact of climate shocks on output along food supply chains.

Common to much of my work is tracing where a shock ends up, through the institutions that stand between it and the households, firms or governments that absorb it. For instance, while higher world oil prices reduce the emissions of economies that import fuels, the effect on the emissions of economies that export fuels depends on how the resulting revenue is spent by their governments. Likewise, the effect of a drought on a household's consumption depends on whether the drought has also affected its lender or relatives. Often, I study how a shock propagates through an institution by exploiting the fact that the institution is exposed to a common shock to a predetermined degree. In other work, I use information from financial markets or institutions to separate the unexpected part of a shock from its expected part, or I compare how the same household responds to shocks that differ in whether they also hit the people around it. Across these papers, I have mostly used hand-constructed panel datasets, and I can provide replication scripts that produce the main results from the raw data.

Fossil revenue, carbon pricing and lock-in. *Where fossil windfalls go.* Higher fuel prices and a higher price of carbon reduce emissions and economic activity in importing economies (Metcalf and Stock 2023; Känzig 2026). A large literature has studied how economies respond to oil price shocks, on both the importing and the exporting side, as world oil prices fluctuate (e.g. Kilian 2009; Baumeister and Hamilton 2019; Arezki, Ramey and Sheng 2017). Meanwhile, resource economists have developed a set of recommendations for how resource revenue windfalls should be used, and have found that revenue is often spent procyclically (van der Ploeg and Venables 2011; Frankel, Végh and Vuletin 2013). At the same time, a growing literature in climate economics has argued that investments in long-lived capital today can lock in high emissions in the future (Davis, Caldeira and Matthews 2010; Seto et al. 2016). Despite these related literatures, little is known about the effect of fossil fuel revenues on carbon emissions in producer economies. In my job market paper, "Where the Windfall Goes: Fossil Revenue, Domestic Absorption and Carbon Emissions" [1], I study how fossil fuel revenues affect carbon emissions where they are earned. I construct a dataset of fossil fuel revenue windfalls for 163 countries between 1981 and 2024.

The windfall is a shift-share shock, constructed by multiplying countries' predetermined fossil fuel rents by the unexpected part of real world oil, gas and coal prices, so that countries with greater and smaller exposure are compared across years with different price surprises (Goldsmith-Pinkham, Sorkin and Swift 2020; Borusyak, Hull and Jaravel 2022). Since world prices also respond to global demand, I instrument the windfalls using news about oil supply from OPEC announcements (Känzig 2021). I then study the effect of the windfall on government revenues, national accounts and emissions in every sector of the EDGAR database, using local projections (Jordà 2005).

I find that, in the 75 economies for which I have data on resource revenues, windfalls enter government budgets almost fully, are saved mostly abroad in the first year, and are spent on consumption, investment and imports within two years. Moreover, I find that each dollar of windfall revenue increases domestic CO_2 emissions by 0.52 kg (with a weak-instrument-robust interval of 0.20 to 0.89 kg), roughly half the carbon intensity of a typical dollar of GDP. The lower carbon intensity is explained by the fact that much of the revenue is saved or spent on imports. The increased emissions mostly come from road transport, power generation and buildings, and the estimated carbon content of windfalls matches that implied by independently compiled energy statistics. I also find that the increase in cement emissions comes from higher utilisation of existing cement kilns, rather than from the construction of new ones, as would be predicted by putty-clay models of capital. In addition, the effect of revenue windfalls on emissions is asymmetric: positive windfalls increase emissions in cement, buildings and power, while equally sized negative windfalls do not decrease them. I estimate that windfalls during the commodity price boom between 2003 and 2014 led to about 2.7 billion tonnes of additional CO_2 emissions, worth about 500 billion dollars in climate damages at the social cost of carbon. In sum, part of the emission reductions induced by high fuel prices on the importing side is offset on the exporting side, through government budgets, and the size of that offset depends on fiscal institutions.

Why carbon is underpriced, and who pays. Fossil fuel subsidies are among the largest fiscal transfers worldwide (Coady et al. 2017), and the ability to tax is a central state function (Besley and Persson 2009). In "Fiscal Capacity, Not Climate Exposure" [3] (with Olga Gorelkina and Amine Ben Amar), we construct effective carbon prices for 174 countries over 2015 to 2022 and show that only three countries (Sweden, Norway and Finland) price carbon above zero once subsidies are netted out. We decompose the underpricing into a domestic wedge (supply costs, forgone consumption tax revenue and local air pollution, all borne by the underpricing country itself) and a global climate wedge. Fiscal capacity (non-resource tax revenue) predicts the domestic wedge (around two thirds of the shortfall), whereas no observable state characteristic predicts the global climate wedge. Climate vulnerability does not directly increase carbon underpricing, but it is so

closely tied to low fiscal capacity that the burden relative to income is around twenty times as high in the most vulnerable fifth of countries. Most of the underpricing can therefore be addressed by strengthening domestic revenue; the climate wedge requires international coordination. In a companion paper [5], I use these underpricing estimates alongside data on global value chains for 77 countries to show that carbon-intensive sectors make up a larger share of manufacturing value-chain output in disaster years in countries where carbon is underpriced, especially in resource-rich countries with low democratic accountability. The main variation in that paper is across countries, while my job market paper uses variation in fossil fuel windfalls over time for identification.

Carbon prices at the border. The European Union's Carbon Border Adjustment Mechanism (CBAM) applies to emerging economies via their exporters. In current work with Luan Santos [7], we compute the CBAM liability for each of the 1,760 tariff codes in African exports to the EU, using customs declarations, official default values and the statutory phase-in of free allocation. We find that African exporters will face a bill of roughly 2.19 billion euros annually once the mechanism is fully phased in, and that the charge as a share of customs value reaches 244 per cent for cement clinker; because the carbon sits upstream, the most processed goods carry the lowest charge in every sector, so the mechanism does not reconstruct the tariff escalation that trade liberalisation had removed. In case studies of five installations, the ability to verify emissions at the system boundary priced by the CBAM does not appear to be closely related to ownership, size or country income.

Climate risk on financial balance sheets. Another strand of my work examines the pricing and insurance of climate risk by financial markets and households, and the limits of that insurance.

Sovereign credit. Do sovereign credit markets price an economy's physical exposure to climate change or its ability to adapt to that exposure? In "Riding the Climate Wave" [2], with Ben Amar and Gorelkina, we measure the co-movement of sovereign credit risk and climate risk for 59 economies over 2008 to 2021, using wavelet coherence tested against a red-noise null hypothesis. Climate vulnerability and credit default swap spreads move together on short, medium and long time scales. Whereas the gradient of vulnerability across countries disappears when we control for the level of development, we find a strong negative relationship between an economy's climate readiness and its spreads, net of income, debt and governance variables. By this estimate, emerging and developing economies with the readiness of advanced economies tend to have spreads about 74 per cent smaller. Capital markets thus appear to reward the ability to adapt, while giving little weight to exposure once development is accounted for. While revising the paper, I discovered that the conventional way of summarising wavelet coherence overstates it, corrected the measure and withdrew a claim that had rested on it.

Households. In poor economies, households have elaborate mechanisms for insuring idiosyncratic risks, but these are generally local, and so they offer little protection against covariate risks such as droughts (Townsend 1994). Support from relatives living far away can fill part of that gap (Rosenzweig and Stark 1989; Yang and Choi 2007). In "Financial Reach and Household Coping under Climate Shocks" [4], with Todd Cort, we use harmonised shock and coping modules covering ten African countries to look at within-household differences in the response to a covariate climate shock and an idiosyncratic shock. We find that the burden of a covariate shock falls largely on the household's consumption, and that the shift towards reducing consumption is 4.6 percentage points smaller among households that receive support from outside their local area, defined as elsewhere in the country. Access to credit, by contrast, makes no measurable difference, because the local credit provider is also affected by the covariate shock. Rainfall in the growing season remains correlated up to about 254 km away, which gives a sense of the distance that support must travel to escape the shock. Simply owning a financial instrument may therefore not make a household resilient if the income stream backing that instrument is exposed to the same shock.

Ongoing work. With Luan Santos, I am assembling monthly data on emerging market corporate bonds, together with sovereign spreads, rating actions, disaster records and measured tracks of cyclones. Our pre-registered tests so far indicate no transmission of sovereign borrowing costs to corporate borrowing costs in the wake of disasters, nor any sign that state backing shelters firms. We will report these null results fully in the paper [9].

Climate shocks along food supply chains. Most studies of the impact of climate change on macroeconomic outcomes treat the food system as a single sector, which implies that a weather shock affects farmers and food manufacturers in the same way. In "Heat beyond the Farm Gate" [6], with Ben Amar and Gorelkina, we estimate the monthly response of the food and drink industries in 21 European countries to local weather conditions over 2000 to 2026, identified from differences in weather across countries in the same month. A summer month 1 °C above normal decreases raw milk deliveries by 0.3 per cent and bakery output by 0.8 per cent, but increases beverage output by 2.0 per cent. Losses in milk and meat are linked to days above 30 °C and are not made up in later months, consistent with heat stress on livestock, whereas beverage output tracks soft drink prices, consistent with a shift in demand. Because 81 per cent of the gross value added response to heat is offset by the opposite response in other parts of the food system, the response of the aggregate food sector understates the exposure of the livestock-based chain. In related work on the United Kingdom [8], a Monte Carlo experiment shows that assuming a single climate response for the whole food sector eliminates the persistent decline in output, which means that the implied climate parameter depends on the sectoral composition of the food system and is not a structural parameter.

In future work, I will keep tracing where shocks end up. I hope to use variation in the adoption of fiscal institutions (fiscal rules, sovereign wealth funds) to identify which institutions reduce the carbon locked in after windfalls, using satellite data on plant activity, which I have validated against national cement output. I also hope to measure the extent to which sovereign and corporate debt markets price the transition risk faced by fossil fuel producers, as their exports meet carbon prices at foreign borders and the rents from their fossil fuel assets shrink. On firms and households, I want to understand whether the financial support open to them survives a climate shock that strikes everyone around them at once. Before I became an economist, I was trained as a soil scientist and worked on agricultural programmes in Nigeria, an experience that still shapes my research questions.

Papers

[1] Uke, O. D. "Where the Windfall Goes: Fossil Revenue, Domestic Absorption and Carbon Emissions". Job market paper. Submitted, *Environmental and Resource Economics*.

[2] Uke, O. D., A. Ben Amar and O. Gorelkina. "Riding the Climate Wave: A Wavelet Analysis of Sovereign Risk in Developed and Developing Economies". Revise and resubmit, *International Review of Financial Analysis*; revision submitted September 2026.

[3] Uke, O. D., O. Gorelkina and A. Ben Amar. "Fiscal Capacity, Not Climate Exposure: Decomposing Global Fossil-Fuel Underpricing and Its Equity Burden". Submitted, *Energy Policy*.

[4] Uke, O. D. and T. Cort. "Financial Reach and Household Coping under Climate Shocks: Evidence from Ten African Countries". Submitted, *World Development*.

[5] Uke, O. D. "Fossil Fuel Underpricing, Climate Disasters and the Carbon Composition of Manufacturing: Evidence from Global Value Chains". Submitted, *Economic Analysis and Policy*.

[6] Uke, O. D., A. Ben Amar and O. Gorelkina. "Heat beyond the Farm Gate: Offsetting Supply and Demand Responses in European Food and Drink Manufacturing". Working paper, 2026.

[7] Uke, O. D. and L. Santos. "Carbon Accounting Capability and Exposure to the EU Carbon Border Adjustment Mechanism: Evidence from African Exporters". Work in progress.

[8] Uke, O. D., A. Ben Amar and O. Gorelkina. "Climate Shocks and Food Manufacturing: Aggregation Bias in the United Kingdom". Work in progress.

[9] Uke, O. D. and L. Santos. "Sovereign Climate Risk and Corporate Bond Spreads: Evidence from Emerging Markets". Work in progress.

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